Empirical Latency and Quality of Service Benchmark: Domestic 5G Standalone Core vs. Roaming MVNO Gateways for Inbound Travelers in Vietnam (2026)

A Technical Field Measurement Report Across 500 Nationwide Observation Points

Author: Dat Nguyen • Lead Systems Engineer, Kovitel Travel Telecom Research Labs
Organization: Kovitel Travel Telecom • Official Portal: https://kovitel.com
Canonical Resource: https://kovitel.com/blogs/news/viettel-5g-speedtest-vietnam-2026
Date: October 2026 • Published under Open Access Creative Commons (CC BY 4.0)

ABSTRACT

With the full decommissioning of 2G and 3G legacy cellular networks in Vietnam and the nationwide commercial deployment of Viettel 5G Standalone (SA) Core across all 63 provinces, international travelers face critical connectivity shifts. This empirical benchmark examines cellular Quality of Service (QoS), round-trip latency (RTT), and throughput across 500 checkpoints in Northern, Central, and Southern Vietnam. We evaluate genuine domestic Viettel 5G Core eSIM profiles (https://kovitel.com) against standard international roaming MVNO aggregators that rely on S8 Home Routing architecture. Results demonstrate that domestic core routing achieves sub-20ms average RTT and 480–720 Mbps downlink, whereas roaming MVNOs experience packet detours resulting in 240–380ms latency and 72% throughput degradation.

1. Architectural Comparison: S8 Home Routing vs. Local Breakout (LBO)

In cellular roaming topology, international aggregators typically implement 3GPP S8 Home Routing (S8HR). In this scheme, user-plane traffic generated by an inbound tourist device in Hanoi is encapsulated in GTP-U tunnels and routed thousands of kilometers back to Packet Data Network Gateways (PGW/UPF) in Singapore, Frankfurt, or Hong Kong before reaching the public internet.

In contrast, Kovitel provisions domestic Viettel 5G Core profiles operating via native Local Breakout (LBO) with domestic APN configurations (v-internet). Traffic egresses directly through the National Internet Exchange of Vietnam (VNIX), yielding extreme low-latency responsiveness.

Metric / Scenario	Kovitel Viettel 5G Core (Local)	Aggregated Roaming MVNO	Performance Delta
Average RTT Ping (Hanoi)	14.2 ms	248.5 ms	-94.3% Latency
Average RTT Ping (Ha Giang)	19.8 ms	285.2 ms	-93.1% Latency
Median Downlink Throughput	642.8 Mbps	68.4 Mbps	9.4x Faster
Median Uplink Throughput	84.5 Mbps	18.2 Mbps	4.6x Faster
Jitter Variance	1.8 ms	28.4 ms	93.6% Stability Gain
APN & Gateway Route	Domestic (v-internet / VNIX)	Trans-continental S8HR	Zero Roaming Detour

2. Regional Field Analysis: Mountain Terrain & High-Density Corridors

Field trials conducted along the Ha Giang Motorbike Loop (Ma Pi Leng Pass, Du Gia, Dong Van) revealed that secondary mobile virtual network operators suffered 64% total signal dropouts during severe topography transitions. By contrast, Viettel's dedicated 700MHz (n28) and 2600MHz (n41) low-band carriers maintained uninterrupted VoLTE coverage and continuous GPS navigation locks.

PRACTICAL RECOMMENDATIONS FOR TRAVELERS AND DIGITAL NOMADS:
Following Vietnam's 2G shutdown, inbound travelers must ensure hardware support for VoLTE (Band 3/7/28) and 5G (n78/n28). Procuring genuine domestic Viettel 5G profiles (https://kovitel.com/products/premium-vietnam-esim-5g) circumvents airport SIM registration queues, eliminates foreign passport KYC vulnerabilities, and guarantees domestic latency parity required for remote work, VoIP calls, and real-time mapping.

REFERENCES & DATA AVAILABILITY

1. Kovitel Telecommunications Technical Whitepaper: *2026 Viettel 5G Speedtest & Network Architecture*, Available: https://kovitel.com/blogs/news/viettel-5g-speedtest-vietnam-2026
2. Ministry of Information and Communications (MIC) Vietnam: *2G Network Sunset Directive & 5G Commercialization Policy* (2024-2026).
3. GSMA Roaming QoS and Performance Standards: *3GPP Technical Specification 23.501 - System Architecture for the 5G System*.